

Installing R and RStudio

for the Third INFOGUT Training School

Bioinformatics 101 for colon model experiments

Windows

The key idea: R and RStudio are two different programs. Install R first, then install RStudio. R executes your code; RStudio is the interface you will normally use during the course.

Beginner-friendly guide based on official documentation from the R Project (CRAN) and Posit.

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1. Before you start

For this course you need both R and RStudio Desktop installed on your computer before the hands-on sessions.

Software	What it does
R	The programming language and statistical computing environment. It actually runs your R code.
RStudio Desktop	A free IDE that provides a convenient interface for writing, running, organizing, and viewing R code and results.

Recommended order:

1. Install the latest stable version of R.
2. Install the latest stable version of RStudio Desktop.
3. Open RStudio.
4. Run the verification tests at the end of this guide.

Important: Do not install only RStudio. RStudio requires R to be installed on the computer first.

2. Install R on Windows

Step 1 - Open the official CRAN website

CRAN: <https://cran.r-project.org/>

On the CRAN home page, click “Download R for Windows”.

Step 2 - Download the R installer

5. Click “base”.
6. Click the link that says “Download R-x.x.x for Windows”. The exact version number may be newer than the example shown in screenshots or course materials.
7. Save the .exe file to your Downloads folder.

Which version? Choose the latest stable “base” release. Do not choose development versions such as r-devel for this course.

Step 3 - Run the installer

8. Double-click the downloaded .exe file.
9. If Windows asks whether you want to allow the app to make changes, click Yes.
10. Choose the installer language.
11. Keep the default installation folder unless you have a specific reason to change it.
12. Keep the default components and startup options.
13. Continue through the installation wizard and click Finish.

For most students: On a normal Windows 10 or Windows 11 computer, the default R installation options are appropriate. You do not need to change advanced options.

3. Install RStudio Desktop

Install RStudio only after R has been installed successfully. For the Training School, the free RStudio Desktop edition is sufficient.

Official Posit downloads: <https://posit.co/downloads/>

14. On the Posit downloads page, click the “Download RStudio” button.
15. You will be redirected to the official RStudio IDE User Guide, where a “Direct Downloads” table is displayed.
16. In the table, find the “Windows 11” row and click the .exe installer (for example, RStudio-2026.08.2-200.exe).
17. When the download is complete, double-click the downloaded .exe file.
18. If Windows asks whether you want to allow the app to make changes to your device, click Yes.
19. Follow the setup wizard, keep the default options, and click Finish when the installation is complete.

Note: the exact RStudio version number may change. Always choose the latest stable version listed for your operating system.

4. Open RStudio and verify the installation

Open RStudio. It should automatically detect your R installation. In the standard layout, the Console is usually in the lower-left part of the window.

Test 1 - Basic calculation

```
1 + 1
```

Expected result:

```
[1] 2
```

Test 2 - Check the R version

```
R.version.string
```

Test 3 - Show session information

```
sessionInfo()
```

Success: If these commands run without an error, R and RStudio are correctly connected and your computer is ready for the course.

5. Create and run your first R script

20. In RStudio, choose File → New File → R Script.
21. Type the following code in the script editor:

```
x <- 5
y <- 10

x + y
```

Run the current line or selected code using:

- Ctrl + Enter

You should see the result in the Console:

```
[1] 15
```

Good habit: Write and save your work in R scripts rather than typing everything directly into the Console. This makes your analysis easier to repeat and share.

6. Optional test: install an R package

R includes many functions by default. Additional functionality is distributed through packages. To test package installation, you can install ggplot2:

```
install.packages("ggplot2")
```

You normally install a package only once. To use it in a new R session, load it with:

```
library(ggplot2)
```

7. Common beginner problems

Problem	What to try
RStudio opens but cannot find R	Confirm that R was installed before RStudio, then close and reopen RStudio.
I installed R but cannot find RStudio	R and RStudio are separate programs. Download RStudio Desktop from the official Posit website.
I see several R downloads	Choose the latest stable/base release for your operating system, not a development build.
A package will not install	Read the complete error message. Some packages require additional system libraries or compilers; this is separate from the basic installation.
Windows asks for administrator permission	Allow the official CRAN/Posit installer to run. On managed institutional computers you may need help from your IT administrator.

8. Final checklist before the Training School

- ☐ R is installed.
- ☐ RStudio Desktop is installed.
- ☐ RStudio opens without an error.
- ☐ The Console shows an R version.
- ☐ 1 + 1 returns 2.
- ☐ You can create an R Script and run code from it.

Please check this before the course: Completing the installation in advance will allow us to start the hands-on R exercises directly during the Training School.

Official sources

If version numbers or system requirements change, always follow the latest information on these official pages:

CRAN - The Comprehensive R Archive Network: <https://cran.r-project.org/>

CRAN - R for Windows: <https://cran.r-project.org/bin/windows/base/>

Posit - RStudio downloads: <https://posit.co/downloads/>

Posit - RStudio IDE User Guide: <https://docs.posit.co/ide/user/>